AquaWatch Rwanda: Rural Water Point Reliability and Maintenance Prioritization System

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Problem Statement

In 2023/24, 87% of rural Rwandan households used an improved drinking-water source, yet non-functioning sources were the most common reason for not using the nearest source (45%), reaching 60% in Eastern Province (National Institute of Statistics of Rwanda [NISR], 2025). Just under 60% of rural piped facilities are operational, largely because of weak operation and maintenance (Japan International Cooperation Agency [JICA], 2025). Evidence from nine countries shows 79% of older rural water points function and that piped tap stands fail more often than hand-pump boreholes (Murray et al., 2024), while functionality also depends on local institutions and governance (Whaley & Cleaver, 2017). Rwanda already runs a WASH management information system and a WASAC GIS inventory of rural systems (Ministry of Infrastructure, 2020; WASAC RWSS Department, n.d.), but public documentation does not describe a transparent, rule-based method for ranking individual water points for maintenance with an audit trail.

Proposed Solution

AquaWatch Rwanda is an Oracle database with PL/SQL that organizes water points, locations, technologies, functionality observations, inspections and maintenance work. It keeps two layers apart: original survey observations (read-only history) and new inspection and maintenance records created through procedures. Documented rules classify each water point as Critical, High, Medium or Low priority using functionality status, days since last inspection, repeat failures and technology type. The system is a prototype for the sampled districts and does not represent current national conditions.

Data Source

Water For People Rwanda 2021 from the Water Point Data Exchange: 2,057 valid records from sampled districts, reported January 2015 to June 2021 (Water Point Data Exchange, 2021). WPDx states that providers license data under CC BY 4.0, so attribution will be given. Data will be inspected for columns, personal information and completeness; 77 rows with unusable status will be stored as Unknown, not guessed. Inspection and maintenance records are system-generated, not survey data.

Planned Tables

DISTRICTS, SECTORS, WATER_POINT_TYPES, WATER_POINTS, FUNCTIONALITY_STATUSES, STATUS_OBSERVATIONS, INSPECTIONS, MAINTENANCE_RECORDS, TECHNICIANS, APP_USERS and AUDIT_LOG. A district has many sectors, a sector many water points, and each water point many observations, inspections and maintenance records. Final keys follow 3NF normalization in Phase II.

Planned PL/SQL Features

Procedures record_inspection, open_maintenance and close_maintenance save work and update status. Functions fn_priority_score and fn_functionality_rate compute priority and district reliability. An explicit cursor builds the overdue-inspection alert list. Package PKG_AQUAWATCH groups this logic. Triggers block maintenance on a decommissioned water point, stop a work order closing before it

opened, write all changes to AUDIT_LOG, and deny master-data changes in restricted periods. Custom exceptions return clear errors.

Innovation

Option: User interface. Tool: Python Streamlit with python-oracledb, connected live to the Oracle database. It shows district comparisons, maps and priority lists, and records inspections by calling the PL/SQL procedures. All rules stay in PL/SQL.

References

Japan International Cooperation Agency. (2025). *JICA project brief note: The project for rural water supply services and infrastructure management development in Rwanda*. https://www.jica.go.jp/oda/project/1904443/__icsFiles/afieldfile/2025/12/02/briefnote_02_en_1.pdf

Ministry of Infrastructure. (2020). *Concept note for WASH MIS data collection in Gicumbi district*. Republic of Rwanda. https://ircwash.org/node/87182

Murray, A. L., Stone, G., Yang, A. R., Lawrence, N. F., Matthews, H., & Kayser, G. L. (2024). Rural water point functionality estimates and associations: Evidence from nine countries in sub-Saharan Africa and South Asia. *Water Resources Research, 60*(2), e2023WR034679. https://doi.org/10.1029/2023WR034679

National Institute of Statistics of Rwanda. (2025). *EICV7 thematic report: Utilities and amenities 2023/24*. https://www.statistics.gov.rw/integrated-household-living-conditions-survey-7-eicv-7/eicv7-thematic-report-utilities-and-amenities-202324

Oracle. (2024). *Oracle Database PL/SQL language reference* (21c). https://docs.oracle.com/en/database/oracle/oracle-database/21/lnpls/

Rural Water and Sanitation Services Department, Water and Sanitation Corporation. (n.d.). *Data collection by QGIS/QField for O&M work of rural water supply systems in Rwanda*. https://qfield.org/docs/case-studies/rwanda-rural-water.html

Streamlit. (n.d.). *Streamlit documentation*. https://docs.streamlit.io/

Water Point Data Exchange. (2021). *Water For People Rwanda 2021* [Data set]. https://catalog.waterpointdata.org/datasets/wfp-rwanda-621f8339/index.html

Whaley, L., & Cleaver, F. (2017). Can 'functionality' save the community management model of rural water supply? *Water Resources and Rural Development, 9*, 56–66. https://doi.org/10.1016/j.wrr.2017.04.001